From Prohibition to Architecture

Artificial Intelligence and the Extended Phenotype of *Ribā* in Financial Systems

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Abstract
The prohibition of *ribā* is often described either as a religious rule against interest or as a formal constraint around which economically equivalent credit products are engineered. Both descriptions are incomplete. A prohibition that persists across centuries does more than invalidate clauses. It changes which contracts are available, which authorities certify them, how distress and default are governed, and which distinctions financial institutions must preserve. Artificial intelligence introduces a new layer to this architecture. When plural and context-dependent legal judgments are translated into taxonomies, retrieval systems, classifiers, and audit rules, some interpretations become easier to observe, reproduce, and scale than others. This article develops a bounded extended-phenotype account of that process. It does not claim that Islamic law is homogeneous, that *ribā* is reducible to every modern interest rate, or that AI has already changed doctrine. It argues that computational legibility can become a selection pressure within Sharia governance: interpretations and products that can be encoded, audited, and explained at scale may acquire an operational advantage independent of their substantive jurisprudential merit. The argument is tested against organized *tawarruq*, *sukūk* standardization, debt restructuring, comparative banking evidence, and recent proposals for AI-assisted Sharia auditing. The article distinguishes documented institutional effects from unverified technological claims, compares the framework with path dependence and the sociology of classification, and states falsifiers. Its practical implication is narrow. AI may support retrieval, anomaly detection, and evidentiary organization, but any system capable of influencing Sharia compliance should preserve provenance, jurisdiction, doctrinal disagreement, uncertainty, abstention, and meaningful human authority.

Keywords: *ribā*; Islamic finance; Islamic banking; artificial intelligence; Sharia governance; extended phenotype; computational legibility; *tawarruq*; *sukūk*; AAOIFI; legal evolution; legal transplants.
JEL classification: K20; Z12; G21; O33; B52.

Transliteration note: technical transliteration is retained for specific Arabic legal terms; conventional English forms are used for names and expressions established in the English-language sources cited.

1. Introduction

A prohibition can fail in more than one way. It can be ignored. It can be enforced so rigidly that the activity moves elsewhere. Or it can remain formally intact while the economic function it constrains returns in another contractual form. The history of credit contains all three responses. Rules against usury did not abolish the demand for liquidity, the price of delay, or the problem of default. They altered the legal forms through which those pressures were expressed and the institutions authorized to distinguish a lawful return from an unlawful increment (Homer and Sylla 2005; Reed and Bekar 2003).

Islamic finance makes this institutional problem unusually visible. The Qur'anic prohibition of *ribā* is not a technical drafting preference. It belongs to a normative order that also addresses trade, uncertainty, gambling, unjust enrichment, hardship, and protection of wealth. Its contemporary implementation nevertheless occurs through banks, capital markets, regulators, courts, Sharia boards, accounting standards, and products designed to compete with conventional finance. Those institutions must translate jurisprudential distinctions into repeatable transactions. The translation is neither uniform nor politically neutral. It distributes authority, cost, and risk.

Artificial intelligence adds a second translation. A Sharia board may reason from text, precedent, commercial purpose, local regulation, and a school-specific methodology. A computational system requires sources, categories, features, labels, and stopping rules. It must decide which version of an authority applies, whether disagreement is represented as an alternative or an error, and when uncertainty should prevent an automated conclusion. These design choices can affect practice even when the system is formally described as decision support.

This article asks: **under what conditions can AI-assisted Sharia governance change which interpretations of *ribā* become operationally legible, auditable, and scalable, and how could that claim be falsified?** The answer is deliberately limited. Current literature does not demonstrate that AI has altered Islamic doctrine, caused convergence among Sharia boards, or improved the systemic performance of Islamic finance. The recent AI and Sharia-governance literature reviewed for this article is doctrinal, conceptual, or based on perceptions rather than independent technical audits (Azam et al. 2026). The stronger causal claim must therefore remain a research hypothesis.

The article advances a narrower proposition. The prohibition of *ribā*, as interpreted through identifiable institutions, can produce external and persistent effects in contracts, governance bodies, standards, and data practices. In that bounded sense, these arrangements may be studied as an extended institutional phenotype. When AI mediates access to authorities and screens transactions, computational legibility becomes one possible selection pressure within that phenotype. A rule or product need not be jurisprudentially superior to gain an operational advantage. It may spread because it is easier to encode, verify, benchmark, or defend in an audit.

This formulation avoids three errors. First, it does not treat Islam or Sharia as a single decision-maker. Islamic jurisprudence contains multiple schools, methods, institutions, and contemporary disagreements. Second, it does not equate *ribā* mechanically with every economic return or use "interest" as a complete translation of a contested doctrinal category. Third, it does not infer substantive justice from formal Sharia certification. Organized *tawarruq* illustrates why form, economic effect, and jurisprudential acceptance must be kept separate: contemporary practice can reproduce the cash-flow profile of interest-bearing credit while remaining contested by authoritative bodies (Ahmed and Aleshaikh 2014; International Islamic Fiqh Academy 2009).

The political implication is that a system making one interpretation easier to scale can redistribute authority even when no formal rule delegates authority to the system.

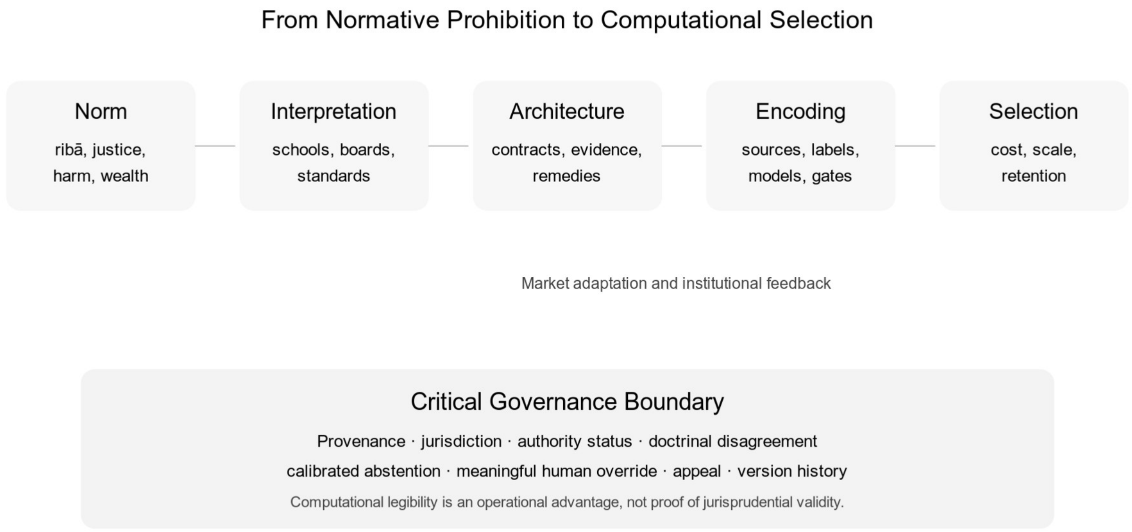

Figure 1. From normative prohibition to computational selection. Author's elaboration.

2. Doctrinal Boundaries and Method

Epistemological positioning. This article treats the extended-phenotype account and computational-legibility mechanism as components of a falsifiable research programme rather than established empirical laws. Following Popper (1959) and Lakatos (1978), adverse evidence must restrict or displace the framework rather than be reclassified as hidden confirmation. Campbell (1974) informs the selectionist vocabulary, but no biological identity between genes and legal rules is asserted. Empirical claims are attributed to their original sources. Proposed mechanisms and predictions remain unvalidated until independently tested.

2.1 What this article does not decide

This article does not offer religious rulings or determine whether a particular financial product is permissible. Nor does it purport to resolve the full relationship between *ribā al-nasīʾa*, *ribā al-faḍl*, modern bank interest, profit, rent, sale on deferred terms, or compensation for actual loss. Those questions require attention to sources, interpretive method, transaction structure, jurisdiction, and appropriate scholarly and institutional expertise.

The Qur'an condemns *ribā* and distinguishes it from trade, including in verses 2:275–279 (Abdel Haleem 2004). Modern writers such as Chapra (2008) defend a broad prohibition of predetermined increments on loans and connect it to justice and risk sharing. That position has influenced contemporary Islamic finance, but an academic account must distinguish the existence of a prohibition from agreement over the characterization of every modern instrument. The scope of *ribā*, the legal causes applied to exchange, the treatment of organized commodity transactions, and the relationship between form and purpose continue to generate disagreement.

For that reason, "the prohibition of *ribā*" in this article means neither an unchanging rule with one self-evident application nor the preferences of a single standards body. It denotes a family of normatively related prohibitions and interpretive practices carried through specific jurisprudential and institutional lineages. Claims are attributed to those lineages whenever the available evidence permits. Where it does not, the article states the limit rather than inventing consensus.

2.2 Genre and evidentiary standard

The article is a theoretical-architecture paper. It synthesizes documented patterns to propose a causal mechanism and testable predictions. It does not report an experiment, field study, or deployment. The three recent papers supplied for this inquiry examine good faith in successive restructuring, *taʿassur* in Indonesian insolvency, and AI-assisted Sharia governance (Marwah et al. 2026; Dahlan et al. 2026; Azam et al. 2026). They are relevant legal analyses. They do not establish causal effects of Islamic doctrine or measured performance of AI systems.

Four evidentiary levels are therefore separated throughout:

1. **Primary or institutional authority**, such as official resolutions, standards drafts, or legislation.
2. **Empirical evidence**, such as comparative bank-level studies.
3. **Doctrinal and historical analysis**, which supports interpretation or institutional genealogy but not causal performance claims.
4. **Model-based prediction**, introduced by this article and awaiting independent testing.

This hierarchy is not a ranking of religious authority. It is a discipline for academic inference. An official resolution can establish what an institution decided; it cannot by itself show how markets changed. A bank-level study can identify associations within its sample; it cannot decide whether a product complies with Sharia.

2.3 Method

The research combines doctrinal analysis, controlled historical comparison, institutional mechanism design, and adversarial literature review. Sources were selected to include: scholarship on the nature of *ribā*; official treatment of organized *tawarruq*; critiques of form and substance; long-run histories of interest; comparative studies of Islamic and conventional banks; official stability reporting; the three recent legal papers; and work on classification, path dependence, and extended phenotypes.

The comparison with Christian usury is controlled rather than genealogical. It does not assume that Christian and Islamic prohibitions share the same sources, content, authority, or trajectory. It asks a narrower question: what institutional responses recur when a durable normative order restricts returns on loans while demand for credit persists? Similar responses can reveal a general mechanism without erasing doctrinal difference.

3. Interest, Prohibition, and Institutional Adaptation

3.1 The price of credit precedes modern banking

Homer and Sylla (2005) document rates on loans across millennia, beginning with ancient Mesopotamia. Their evidence is uneven, as they repeatedly acknowledge, because loan types, maturities, enforcement, currency, and surviving records are not directly comparable. The durable point is not a universal rate series. It is that societies priced delay and default risk long before modern banking and did so through institutions shaped by money, security, trade, political order, and law.

Restrictions did not operate in an empty economy. Borrowers still needed consumption smoothing and working capital. Merchants needed settlement across time and distance. States needed funds. When direct interest was condemned or capped, financial practice adapted through permitted forms, tolerated exceptions, disguised returns, specialized intermediaries, or migration outside the regulated community.

The Christian history of usury illustrates this effect. Medieval doctrine could approach a general prohibition, yet commercial expansion generated debates over partnership, rent, exchange, compensation for loss, and titles extrinsic to the loan. Reed and Bekar (2003) show that religious prohibitions against usury changed over time and across traditions. Homer and Sylla describe the resulting effect on European credit forms. The prohibition mattered even when evaded because actors spent resources to classify transactions, obtain authorization, or shift to a permissible form.

3.2 Why the historical analogy must remain controlled

The analogy cannot prove the meaning or consequences of *ribā*. Christian doctrine developed through different texts, ecclesiastical structures, state relations, and economic transformations. Islamic commercial law also developed institutions of partnership, sale, lease, and risk allocation that cannot be reduced to evasions of a loan prohibition. Treating every alternative contract as hidden interest would beg the question the analysis is supposed to investigate.

The historical comparison supports only a structural proposition: a durable prohibition can shape the design space even when the constrained economic function persists. It can change transaction costs, create demand for interpretive expertise, and select forms that survive both legal scrutiny and market competition. Whether a particular Islamic product represents genuine risk sharing, lawful sale, necessary accommodation, or impermissible circumvention remains a product-specific and authority-specific question.

3.3 From price suppression to architectural selection

The phrase "prohibition of interest" suggests a binary outcome: an increment is present or absent. Institutional practice is multidimensional. A transaction allocates ownership, possession, market risk, credit risk, delay, remedies, collateral, and losses. Two transactions can produce similar expected cash flows while differing in property rights or consequences under stress. Conversely, different legal forms can converge so closely in economic operation that form-substance criticism becomes unavoidable.

This is the first architectural effect. A prohibition selects not only a number but a bundle of permissible relations. It also selects verification institutions. Someone must determine whether ownership transferred, whether the asset existed, whether the sequence was prearranged, whether uncertainty was excessive, and whether a hardship remedy merely capitalized an unlawful increment. The norm therefore extends into evidence, procedure, and organization.

4. The Extended Institutional Phenotype of *Ribā*

4.1 A bounded use of extended phenotype theory

Dawkins (1982) used the extended phenotype to direct attention toward effects of replicators beyond an organism's body. That structure has been applied to legal rules and constructed institutional niches (Lerer 2026a, 2026b). The concept should not be imported into law as a decorative synonym for "consequences." A legal application requires an identifiable causal structure: variation among interpretations or institutional implementations; persistence beyond the immediate decision; transmission or copying; expression in behavior; and differential retention.

Let n denote a normatively specified interpretation of *ribā*. Let an institutional environment at time t be:

$E_t = (J_t, R_t, M_t, C_t, A_t, D_t)$,

where J represents jurisprudential authorities, R positive regulation, M market conditions, C available contracts, A audit and certification institutions, and D data and documentation practices.

The observable institutional expression of n is:

$P_t(n) = F(n, E_t)$.

$P_t(n)$ can include approved structures, prohibited clauses, governance bodies, evidentiary requirements, restructuring rules, and audit procedures. The expression is not caused by doctrine alone. Market competition, state regulation, accounting, technology, and organizational incentives affect the phenotype. This is why the model is interactionist rather than essentialist.

4.2 Replication without theological reduction

The unit that persists is not "Islam" and not a divine command treated as a biological gene. The empirical candidate is an institutionalized interpretation: a standard, opinion, contract template, rule of review, or classification that can be reproduced across transactions and organizations. It is copied with varying fidelity, sometimes through formal adoption and sometimes through professional practice.

Differential retention occurs when some implementations spread while others remain local or disappear. Market demand can favor familiar cash flows. Regulators can favor standardized products. Auditors can favor forms with clear documentation. Sharia boards can reject a structure on substantive grounds. None of these selection environments is normatively final. Survival shows fit with a regime, not truth or justice.

4.3 Multilevel selection and authority

Selection occurs at several levels. A bank chooses products that attract customers and satisfy its board. A jurisdiction chooses standards that support market development and legitimacy. An international body promotes harmonization. A jurisprudential community preserves or challenges interpretive lineages. These levels can conflict.

A product may succeed commercially while attracting jurisprudential criticism. A strict standard may preserve a substantive distinction while reducing liquidity or increasing issuance costs. A local accommodation may improve access but weaken cross-border recognition. The model therefore rejects a single fitness function. Institutional success, commercial success, regulatory acceptance, and jurisprudential validity are separate outcomes.

5. Computational Legibility as a Selection Pressure

5.1 From judgment to data

AI systems do not consume doctrine directly. They operate on representations. Texts are digitized and segmented. Authorities receive metadata. Transactions become fields and events. Outcomes become labels. Exceptions become rules, examples, or residual uncertainty. The representation determines what the system can distinguish.

Bowker and Star (1999) showed that classification systems are infrastructural: categories organize work and distribute visibility. Espeland and Stevens (2008) described how quantification transforms the objects it measures. These insights matter to Sharia governance because computational support can make one interpretation cheap to retrieve and another costly to represent. A system can preserve formal pluralism while operationally routing almost every case through one standardized ontology.

Define the computational legibility of interpretation i within system s as a vector:

$L(i,s) = (d, r, v, x, a)$,

where d is documentary availability, r representability in the system's schema, v verifiability against accessible evidence, x explainability to reviewers, and a applicability under encoded jurisdictional rules. This is not a validated index. It identifies dimensions for measurement.

An illustrative application shows what the vector contributes. Consider two hypothetical review files, without treating either file description as a ruling on permissibility. File A records a commodity-based liquidity transaction through standardized documents but omits reliable evidence of possession, agency sequence, and sale to an independent third party. Its documentary availability and schema representability may be high, while transaction-level verifiability and applicability under the relevant authority remain low or unresolved. File B records a sale-based financing transaction with identified assets, timestamped title transfer, disclosed agency, and jurisdiction-specific authority metadata. Its five dimensions may all be comparatively high. The contrast exposes a central point: machine legibility can reward documentary regularity even when the legally material facts are incomplete. L(i,s) must therefore be assessed dimension by dimension rather than collapsed into an unvalidated aggregate score.

The selection hypothesis is:

Pr(adoption or repeated use of i) = G(L(i,s), jurisprudential acceptance, market utility, regulatory fit, organizational cost).

Computational legibility is therefore one influence among several. The hypothesis does not predict that the easiest rule to encode always wins. A respected authority, binding regulation, or substantive objection can dominate. It predicts a measurable advantage when competing interpretations are otherwise viable and institutional workflows reward speed, consistency, and auditability.

5.2 The danger of machine-legible formalism

The distinction between formal compliance and substantive purpose predates AI. Technology can intensify it. Fixed-return patterns, penalty clauses, asset records, and transaction sequences are comparatively tractable. Purpose, coercion, distributive burden, social harm, or commercial substance can require facts beyond the data stream.

A detector optimized on labeled examples may learn the documentary phenotype of prior approvals rather than the underlying jurisprudential reasoning. Institutions then adapt to the detector. Contracts become easier to classify, explanations become templated, and difficult facts migrate outside structured fields. Apparent compliance can rise while substantive scrutiny falls.

This feedback is reflexive:

classification → market adaptation → new observed data → retraining or rule revision → classification.

If the same institution controls product design, data, and evaluation, the loop can entrench a convenient interpretation. If independent authorities preserve dissent and counterexamples, the loop can instead expose recurrent attempts at circumvention.

5.3 Human review is necessary but not self-executing

Azam et al. (2026) correctly argue that AI should support rather than replace human Sharia authority. Yet "human in the loop" describes topology, not effective control. A reviewer cannot exercise authority if the system hides source conflicts, presents an uncalibrated score as fact, or filters out cases before review. Volume and time pressure can turn nominal approval into automation bias.

Meaningful authority requires at least: access to the underlying sources; identification of jurisdiction and interpretive lineage; visibility of competing readings; an abstention mechanism; power to override; a recorded reason; and feedback that does not silently redefine doctrine. The reviewer must be able to reject the system's frame, not merely its answer.

6. Cases of Form, Substance, and Institutional Feedback

6.1 Organized *tawarruq*

Organized *tawarruq* provides the clearest case for the architecture thesis. Ahmed and Aleshaikh (2014) trace historical debate and examine contemporary rulings. They conclude that industry practice can be inconsistent with both historical majority views and contemporary rulings. In its official English version, the International Islamic Fiqh Academy's Resolution No. 179 (5/19) distinguishes classical *tawarruq* from structured and inverse forms and prohibits the latter when coordination produces present cash against a larger future debt (International Islamic Fiqh Academy 2009).

This controversy shows why economic equivalence neither resolves nor dissolves the legal issue. Similarity to an interest-bearing loan supplies a substantive objection, central to broader critiques of form and substance in contemporary Islamic finance (El-Gamal 2006). The transaction sequence, ownership, agency, and coordination supply legal facts. The institutional outcome depends on who classifies those facts and under which methodology.

For computational governance, *tawarruq* creates a demanding test. A system that checks only the presence of commodity trades may certify form. A system that reconstructs agency, prearrangement, possession, and economic effect needs transaction-level provenance across actors. Even then, the final characterization may depend on an authority's position. The correct machine output may therefore be a structured conflict and escalation, not a binary label.

6.2 *Sukūk* and the ownership problem

The distinction between asset-based and asset-backed *sukūk* similarly exposes the relation between form, ownership, and risk. AAOIFI's Exposure Draft of Shari'ah Standard No. 62 on *sukūk* addresses ownership and structural requirements. The latest official AAOIFI statement identified for this article, dated 28 April 2025, described Standard No. 62 as a draft under revision and not finalized. The draft must therefore not be presented here as a uniformly adopted final rule (AAOIFI 2023, 2025).

The draft illustrates institutional feedback: market practice creates concerns; a standards body revisits the architecture; issuers, lawyers, boards, and investors must respond. The systemic consequences cannot be inferred from the draft. Stronger ownership requirements might reduce formalism, but they can also change insolvency treatment, issuance cost, tax consequences, and compatibility with national property law.

An AI system trained on pre-revision documents could preserve an outdated classification. One trained only on the new draft could apply a nonbinding proposal as if it were governing law. Version, authority status, effective date, adoption, and jurisdiction are therefore part of the substantive model, not metadata decoration. The continuing revision process is also a live illustration of P6: two systems using different snapshots of the same authority corpus can return inconsistent results even when neither underlying model has changed.

6.3 Distress, good faith, and restructuring

Marwah et al. (2026) compare successive credit and Sharia-financing restructuring in Indonesia and Japan. They argue for an objective good-faith standard incorporating debtor eligibility, supervision, business monitoring, and restructuring frequency. Their comparison suggests that strict oversight can arise through different normative architectures: Sharia constraints in one setting and feasibility-based supervision in another. The study does not show that religious source alone causes superior outcomes.

Dahlan et al. (2026) use *taʿassur*, or financial hardship, to criticize creditor-initiated Indonesian restructuring that lacks a substantive distress test and sufficient judicial scrutiny. The proposed framework emphasizes capacity, proportionality, feasibility, and harm. Again, the contribution is doctrinal and reform-oriented. Recovery rates and causal effects remain unmeasured.

Together, the papers expand the phenotype beyond origination. The prohibition of unlawful increment interacts with default, forbearance, penalties, creditor power, and the treatment of viable debtors. An AI compliance system that screens only contract formation misses the moment when economic burdens are reallocated through restructuring.

7. What Comparative Banking Evidence Can and Cannot Show

Claims of systemic superiority are particularly dangerous in this field because they can turn religious identity into an empirical shortcut. Available evidence is mixed and context-dependent.

Čihák and Hesse (2008), using banks in eighteen systems, found that small Islamic banks tended to be financially stronger than small commercial banks, while large commercial banks tended to be stronger than large Islamic banks. Beck, Demirgüç-Kunt, and Merrouche (2013) found differences in business orientation and performance but also warned that observed business models may be closer than stylized theory suggests. Hasan and Dridi (2010) found that Islamic banks were affected differently during the global financial crisis: features of their business model limited some first-round effects in 2008, while risk-management weaknesses contributed to larger profitability declines in some settings in 2009. Across the cycle, the comparison was not a simple victory for either model.

The IFSB's 2026 stability report states that the global Islamic financial services industry reached approximately USD 4.4 trillion in total assets in 2025. It also identifies emerging hybrid risks and continuing structural vulnerabilities across evolving Islamic banking practices (IFSB 2026). This current institutional evidence reinforces the need for differentiated analysis by sector, jurisdiction, size, liquidity infrastructure, and product composition.

None of these studies tests computational legibility. They serve two narrower purposes. First, they reject the inference that formal Sharia compliance guarantees systemic stability. Second, they identify receiver variables that an empirical study of AI must control. Size, jurisdiction, concentration, liquidity, and business model can explain outcomes otherwise misattributed to doctrine or technology.

8. A Governance Architecture for AI-Assisted Sharia Review

The responsible architecture is not an automated "*ribā* detector." It is a source- and disagreement-preserving decision-support system.

8.1 Source layer

Each authority should carry provenance, language, issuing body, jurisprudential lineage where applicable, jurisdiction, legal status, publication date, effective date, amendment history, and adoption status. Translations must remain linked to the original. Secondary summaries cannot silently replace primary sources.

8.2 Interpretive layer

The system should represent multiple supported interpretations when they materially differ. It should not average them into a synthetic consensus. A rule graph can identify agreement, conflict, scope, and dependency. Minority status is not error status.

8.3 Transaction layer

The transaction model should include economic sequence, ownership, possession, agency, consideration, timing, remedies, collateral, penalties, and loss allocation. Document labels alone are insufficient. Missing facts should produce NOT_AVAILABLE, not inference disguised as certainty.

8.4 Decision states

Outputs should change workflow state:

- PASS: the transaction is supported under the identified authority and facts, subject to ordinary review.
- ESCALATE: material interpretive conflict, missing evidence, or contextual judgment remains.
- BLOCK: the requested reliance is unsafe because a binding prohibition, unresolved authority, or critical evidentiary failure prevents responsible processing.

These are governance states, not theological judgments generated by the model. Final authority remains with the competent human or institution.

8.5 Evaluation and appeal

A golden set should preserve expert disagreement rather than force majority labels. Evaluation should measure source retrieval, authority-status accuracy, factual completeness, calibrated abstention, and harmful error separately. False approval and unnecessary blockage do not have equal consequences. Every decision should be appealable, and corrections should preserve version history rather than rewrite the record.

9. Rival Explanations

9.1 Path dependence

Path dependence explains how early choices, increasing returns, coordination effects, and switching costs can lock institutions into trajectories (David 1985; Pierson 2000; Hathaway 2001). It may explain much of the persistence described here without evolutionary vocabulary.

The extended-phenotype framework earns its place only if it adds a specific unit and mechanism: which interpretation or procedural artifact is copied, how it is expressed through institutions, what selects it, and how it transfers between levels. If those elements cannot be identified, path dependence is the simpler account and should prevail.

9.2 Classification and quantification

The sociology of classification may offer an even stronger account of AI-mediated change. Categories make some objects visible and others residual; metrics invite adaptation; standards produce commensurability (Bowker and Star 1999; Espeland and Stevens 2008). “Computational legibility” draws directly from this tradition.

The evolutionary contribution lies in connecting classification to differential retention across repeated institutional cycles. Classification describes the infrastructure. Selection analysis asks why one encoded variant spreads, survives challenge, or invades another workflow. The two frameworks are complementary only when that differential process is demonstrated.

9.3 Decoupling and symbolic compliance

Organizational theory predicts that formal structures can become decoupled from practice. If Sharia certification functions mainly as legitimacy while economic behavior remains unchanged, symbolic compliance may explain the evidence better than an extended phenotype. Organized *tawarruq* makes this objection serious.

The architecture thesis survives only if legal form changes behavior somewhere material: ownership, risk, remedies, documentation, governance, cost, or market access. If no such consequence can be measured, the “phenotype” is descriptive excess.

9.4 Political economy

Standards do not evolve in a neutral arena. Banks, states, scholars, professional firms, technology suppliers, and customers have different resources and incentives. A computational standard can concentrate power in organizations able to produce data and satisfy audit requirements. Political economy may explain adoption more directly than doctrinal fidelity.

This objection narrows rather than defeats the model. Market and state power belong inside the selection environment. But the paper must never redescribe power as anonymous evolution. Actors remain responsible for design, adoption, and reliance decisions.

10. Predictions, Falsifiers, and Research Design

10.1 Testable predictions

The computational-legibility hypothesis yields the following predictions, including competing directional outcomes for interpretive diversity:

P1: Contract mix. After adoption of AI-assisted Sharia auditing, institutions will shift toward products with more standardized and machine-verifiable documentation, controlling for regulation and market demand.

P2: Escalation profile. Human review will concentrate in transactions with missing provenance, hybrid structures, conflicting authorities, or purpose-sensitive analysis.

P3a: Interpretive concentration. Systems built on a narrow authority corpus will reduce observed diversity in decisions, even when formal policy permits multiple interpretations.

P3b: Interpretive expansion. Systems that retrieve provenance-bearing sources across languages, jurisdictions, and jurisprudential lineages may increase observed diversity by reducing the search cost of positions previously inaccessible to reviewers.

P4: Documentation adaptation. Product teams will change drafting and data capture to reduce automated flags. Some changes will improve substantive evidence; others will constitute compliance gaming.

P5: Provider power. When ontologies or models are proprietary and reused across institutions, a small number of vendors will influence operational categories without possessing formal jurisprudential authority.

P6: Version sensitivity. Unmanaged authority updates will produce inconsistent results across otherwise similar transactions. Versioned systems with adoption metadata will reduce this failure.

P7: Formalism gap. Accuracy on clause-level labels will overstate accuracy on transaction-level substantive review unless ownership, agency, sequence, and remedies are represented.

10.2 Falsifiers

The strong selection claim should be rejected in a studied context if adoption of AI produces no change exceeding measurement error and a preregistered smallest effect of interest in contract mix, documentation, escalation, interpretive dispersion, review cost, or provider concentration after controlling for confounders. The empirical protocol must justify that threshold from baseline variance, data quality, and institutional relevance rather than select it after observing results. P3a and P3b should be evaluated as competing directional predictions using the diversity of authorities retrieved, cited, discussed, and reflected in decisions. The claim should also be rejected if human authorities routinely override computational representations without measurable anchoring effects.

The extended-phenotype terminology should be abandoned if the study cannot identify a persistent and copied institutional variant, a causal route to behavior, and differential retention. If path dependence or classification theory explains the same observations with fewer assumptions, those frameworks should take priority.

10.3 Mixed-method design

A feasible first study could compare institutions in Indonesia, Malaysia, and Bahrain or the United Arab Emirates. Jurisdiction selection must be justified by regulatory architecture and data access, not by an assumption that each represents Islamic law as a whole.

The unit of analysis would be a product or transaction review, nested within institution and jurisdiction. A longitudinal design would compare pre-adoption and post-adoption periods where an identifiable AI-assisted workflow exists. Comparison institutions without such adoption would reduce, though not eliminate, confounding.

Data would include: versioned Sharia opinions and standards; transaction and contract features; committee outcomes; disagreement and overrides; escalation reason; processing time; review cost; product mix; and system provenance. Confidential material would require institutional approval, minimization, and controlled access.

Expert labeling should begin by measuring inter-rater disagreement. The golden set should contain both consensus and contested cases. A model should not receive credit for reproducing a forced label where qualified experts disagree. Evaluation should report:

- source and authority retrieval accuracy;
- jurisdiction and effective-date accuracy;
- factual completeness;
- agreement conditional on expert consensus;
- preservation of disagreement;
- calibration and abstention;
- false approvals and false blocks;
- override rates and reasons;
- downstream changes in drafting or product design.

Interviews with Sharia board members, compliance officers, regulators, product lawyers, and system designers would identify mechanisms not visible in logs. The study should ask who selected the ontology, whose sources were omitted, how overrides affect later versions, and whether the system changes the questions reviewers see.

11. Discussion

11.1 What legibility does to authority

Computational legibility looks technical because it is implemented through schemas, retrieval, and classifiers. Its political effect lies in agenda control. The system determines which authorities arrive first, which differences appear material, and which cases receive scarce human attention. A model need not produce a formal religious ruling to influence the practical distribution of interpretive authority.

This effect can be beneficial. Better provenance can expose unsupported products, inconsistent treatment, or missing ownership evidence. Continuous monitoring can reveal conduct that periodic document review misses. Search across a large corpus can help a board locate relevant authorities without surrendering judgment.

The same infrastructure can narrow the field. A commercially dominant ontology can make local or minority interpretations expensive to maintain. Audit efficiency can become a reason to suppress disagreement. Institutions can optimize documentation to the system's expectations. The governance question is therefore not whether AI is permitted to "decide." It is whether the surrounding architecture preserves the conditions under which competent authorities can still decide differently.

11.2 The substance of the extended phenotype claim

The prohibition of *ribā* does not float above institutions. It is carried through contracts, professional roles, records, and remedies. These external forms affect people who may never participate in the original interpretive act. A standard drafted in one setting can determine ownership language in another. A transaction template can preserve a distinction across thousands of contracts. An audit model can reproduce a classification across institutions.

That persistence is the useful content of the extended-phenotype claim. It does not imply that doctrine mechanically causes every outcome. It identifies how an interpretation can remain causally active through artifacts and organizations. AI extends the possible range and speed of that expression. It also increases the cost of a classification error because the error can become infrastructural.

11.3 Neutrality and objective analysis

Analytical neutrality does not require treating all claims as equally supported. Organized *tawarruq* is genuinely contested. Claims of general systemic resilience are not supported by a single uniform empirical record.

Current AI literature does not prove operational effectiveness. These are evidence judgments, not religious judgments.

Nor does neutrality require pretending that architecture is apolitical. Decisions about source hierarchy, minority views, ownership evidence, and escalation distribute practical power. The responsible academic position is to make those choices observable and contestable.

12. Limitations

The article does not include primary interviews, proprietary transaction data, or an audited AI deployment. Its predictions remain untested. The recent papers on AI and Sharia governance are useful for identifying proposed functions and governance concerns, but they do not establish technical performance or causal effects.

The analysis is not an exhaustive treatment of Sunni schools, Jaʿfari jurisprudence, regional custom, or every national framework. It is an institutional and regulatory analysis of Sharia governance, not a reconstruction of the classical jurisprudence of *ribā*. No claim about cross-school consensus is made beyond the cited sources. A publication aimed at a specialist Islamic-law journal should receive substantive review from scholars representing more than one jurisprudential tradition.

The historical comparison relies partly on broad secondary histories. It identifies recurring institutional responses but does not establish a direct genealogy between Christian usury rules and Islamic finance. The English-language source base may underrepresent Arabic, Persian, Bahasa Indonesia, and Malay scholarship.

The extended-phenotype model risks redescribing established institutional mechanisms. Its value depends on future evidence that identifies transmissible variants, causal expression, and differential retention. Computational legibility may prove secondary to regulation, market power, or ordinary path dependence.

Finally, even a well-governed decision-support system can create automation bias, confidentiality risk, discriminatory access, and dependence on incomplete data. Source provenance and human review reduce those risks; they do not eliminate them.

13. Conclusion

The prohibition of *ribā* has never operated only at the level of a sentence. It shapes contracts, authorities, evidence, remedies, and market institutions. The persistence of those effects makes an extended institutional phenotype a plausible research object, provided the underlying variants and selection mechanisms can be identified rather than assumed.

Artificial intelligence does not enter this field as a neutral calculator. It requires doctrine to become data and judgment to become workflow. That transformation can improve retrieval, consistency, and monitoring. It can also privilege what is easy to encode, turn an authority draft into an apparent rule, suppress legitimate disagreement, or reward formal compliance detached from economic substance.

The present evidence supports caution rather than a verdict on AI's effect. There is no demonstrated causal chain from AI adoption to doctrinal change or systemic improvement. There is, however, a testable mechanism: computational legibility can alter the relative cost of preserving, auditing, and scaling interpretations. The next step is to measure that mechanism in real institutions while preserving the disagreements that make the inquiry legally and religiously meaningful.

The governing principle follows from the evidence. AI may organize sources and expose anomalies. It should not manufacture consensus. Where authority, context, or material facts remain contested, the system's most accurate answer is an auditable escalation.

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Artificial intelligence tools assisted with source discovery, extraction, comparison, citation checking, conceptual stress-testing, visualization, and drafting. The author defined the research question, selected the theoretical direction, reviewed the evidence boundaries, and retains responsibility for the interpretations, claims, and publication decision. No private client data were used. AI-generated source suggestions were not treated as verified citations without separate checking against identifiable publications or institutional sources.

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